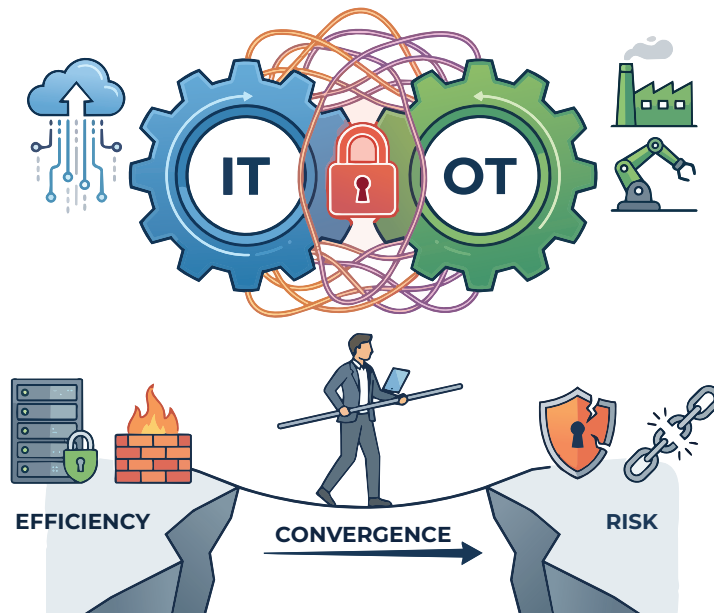


Navigating the Security Paradox of IT/OT Convergence

With physical systems no longer isolated from the digital infrastructure, organisations must adopt new ways of ensuring cybersecurity.



We have witnessed a profound evolution in security over the past few decades, transitioning from the ‘inherently safe’ era of the mainframe to today’s ‘persistently vulnerable’ landscape. Moving from isolated mainframes to distributed client/server models, and then to the web and cloud, we have now arrived in the AI era—where autonomy is the mantra driving business.

Traditionally, OT (operational technology) systems have been air-gapped or isolated. Unlike the digital-only nature of IT, logic executed in an OT environment has a direct, immediate effect on the physical world. Consequently, performance and reliability requirements are significantly more stringent. Because these operations can impact human safety, public health, environmental integrity, and the broader national or

global economy, OT must be viewed through a specialised lens of ‘mission criticality’ that prioritises physical safety and uptime over traditional data-centric IT protocols.

As quantum technology matures and IT/OT convergence becomes seamless, we face critical questions: How will the security landscape pivot? What impact will this have on our critical public infrastructure? And how must our mitigation strategies and regulatory frameworks evolve to keep pace? Monitoring how these emerging trends shape our long-term strategies and collective efforts is the next vital ‘watch-out’ area.

Challenges today

Industrial security is undergoing a seismic shift. Attackers have moved beyond the immediate ‘kill switch’—they are now siphoning operational

telemetry to build digital twins and training AI models on exfiltrated data to orchestrate future, high-precision attacks

Also, there is a widespread belief that data is safe if it is isolated within private networks or completely air gapped. On paper, it sounds reassuring. In reality, it creates a false sense of security. Because isolation is not immunity. Even in air-gapped environments, vulnerabilities exist—and they are often introduced through the very mechanisms designed to operate and maintain these systems:

- **Human factor risk:** Social engineering, credential compromise, or insider threats can bypass even the strongest physical isolation.
- **Data movement channels:** USB drives, external media, and temporary connections for reporting or analytics become silent attack vectors.